FEMU experiment workbook

Use one copy per experiment. This workbook supports the planned tutorial release and the existing canonical walkthroughs. It is not a replacement for the qualified launch/configuration files. Keep actual results, including failures, in a separate attempt directory.

Before running

Research question:

Prediction and why:

Module and tutorial release:

Public FEMU commit and binary hash:

Host/guest kernel, tools and image:

Experimental namespace identity and size:

Why this is not the boot disk or an in-use device:

Configuration and workload hashes:

Initial state and reset procedure:

Record the controlled comparison

Item	Baseline	Changed configuration
Variable intentionally changed		
Variables held fixed		
Exposed namespace bytes		
Raw NAND bytes		
Mapping/GC/placement policy		
Workload requests, byte count and seed		
Conditioning procedure		
Data-verification result		

Item	Baseline	Changed configuration
Exit status and result directory		

Explain any difference not intended by the experiment. If the runs process different request counts or start from different states, decide whether the comparison still answers the original question.

Core checkpoints

S0

Draw the path from the application to completion. Mark where payload data moves and where modeled time is assigned. Identify the mode and source files. Record the verified data size and readback result.

S1

Counter	Before	After	Delta	Unit
Host writes				
NAND writes				
GC writes				
Mapping writes, if applicable				
Padding, if applicable				

State the exact WAF formula used and why its terms cover this configuration. Explain whether GC occurred. A namespace fill is not proof of steady state.

S2

Record the endurance group, placement identifiers, handles, and groups used. Compute interval MBMW/HBMW from byte counters. Describe how you kept the logical request sequence equivalent. Explain a neutral or negative placement result as carefully as an improvement.

S3

Operation	Predicted state/write pointer	Observed state/write pointer	Status
Report initial zone			
Valid write/append			

Operation	Predicted state/write pointer	Observed state/write pointer	Status
Deliberate invalid operation			
Close/finish as assigned			
Reset			

Explain what reset changes and why finish is different. Record data verification before destructive reuse.

S4

Record the input and program hashes, CPU reference output, CSD output, and comparison result. Which bytes crossed the interface? Which clock did you measure? Was `csd_pause` enabled? Does the result establish functional correctness, modeled timing, or real-device performance?

S5

Record the guest kernel configuration, region and access route, verified data, and before/after cache/media counters. Explain why enumeration alone would not prove the workload used the region.

S6

Save the baseline, failing test, patch, and passing test. Explain why the test fails before the fix and what incorrect implementation it would catch. Include the command response schema and a checked invalid-input case.

Interpret and hand over

Observation:

Did it match the prediction? Explain through a state or code path:

Possible confounder and a follow-up experiment that distinguishes it:

What the current model does not represent for this question:

Steps another person needs to reproduce the result:

Independent rerun status and any intervention required:

Completion category: independent / assisted / demonstrated / offline analysis / blocked.

Assessment rubric

Score each criterion 0, 1, or 2: absent/incorrect, partly supported, or complete and supported by the receipt. This rubric evaluates an experiment, not the student's speed.

Criterion	What earns 2
Question and prediction	Specific, testable, tied to the selected model
Control and state	Intended variable isolated; reset and conditioning explicit
Correctness	Device identity and data verification documented
Measurement	Raw evidence, units, interval, formula, and status agree
Interpretation	Explains the mechanism and a meaningful model limitation
Reproduction	Independent person can follow the bundle and obtain the checkpoint

A result need not show a speedup or lower WAF to be a successful experiment. An unexplained improvement is weaker evidence than a well-controlled neutral result with a correct mechanism-level explanation.